Jeffrey Kagle — Portfolio

Selected projects in acoustics, robotics, and product design.

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BIOMECHANICS · 2024 · EMAE 415 BIOMECHANICS · CASE WESTERN RESERVE UNIVERSITY

Biological and Robotic Synthesized Cuttlefish Fin Biomechanical Model

Jeffrey Kagle

Cuttlefish swim by passing a sinusoidal wave along muscular-hydrostat fins, but no experimental study had measured or modeled the thrust these fins produce for forward swimming. This project synthesized biological fin mechanics (anatomy, EMG activation, and force–strain models) with a robotic sinusoidal-fin model into one hybrid force model, creating a benchmark for comparing cuttlefish and designing soft biomimetic robots.

↓ Final Paper

3D
fin force model: dorsoventral, lateromedial, and posteroanterior

2
fin beating modes captured: gentle (aerobic) and vigorous (anaerobic)

11
biological and robotic studies synthesized into one benchmark

Biological fin model

$$p = \frac{F_h}{yt} \qquad F_v = \frac{F_h x^2}{A_0} \qquad F_h = A_{tm} s_{tm}$$

$$F_{v,tot} = 120 T_c \cos(\theta) + 60 T_{dv}$$

$$T = EA\varepsilon$$

$$\varepsilon_c = \sqrt{\frac{\left(\frac{x}{30}^2 + \frac{900 A_0^2}{x^2}\right)}{x_0\sqrt{2}}} - 1$$

$$\varepsilon_{dv} = \frac{A_0}{(x x_0) - 1}$$

Robotic sinusoidal-fin model

$$\begin{cases} F_T(t) = \int_S p(t)(n_x \cdot dS), \\ F_{D(t)} = \mu \int_S \left(\frac{\partial v(t)}{\partial \delta} \cdot n_x \right)\Big|_{\delta=0} dS, \\ \rho \frac{dV}{dt} = F - \nabla p + \mu \nabla^2 V, \\ \nabla \cdot V = 0 \qquad\qquad (i, j, k = x, y, z), \\ \sigma_{ij,i} + F_j = \rho_s \frac{\partial^2 u_j}{\partial t^2}, \\ \varepsilon_{i,j} = \frac{1}{2}(u_{i,j} + u_{j,i}), \\ \varepsilon_{ij} = \frac{1}{E}\left[(1+v)\sigma_{ij} - v\sigma_{kk}\delta_{i,j}. \right. \end{cases}$$

Fig. 1. The two models combined in the hybrid: biological fin-force and strain equations (left, after Johnsen & Kier) and robotic sinusoidal-fin thrust, drag, and fluid–structure equations (right, after Su et al.).

KEY RESULTS

- Built a hybrid thrust model by equating the robotic sinusoidal-fin equations of Su et al. with the biological fin-force model of Johnsen and Kier, assuming a robot of matched size and shape — filling the gap left by the lack of measured fin thrust data for cuttlefish.
- Solved for thrust from the change in velocity over a time segment (acceleration gives thrust minus drag), then computed drag at a point velocity to isolate point thrust and complete the 3D force equations.
- Linked fin physiology to propulsion: muscles fire in a sinusoidal activation wave, gentle beating uses only the aerobic transverse muscle, and vigorous beating recruits all three muscle groups for higher force.
- Included elastic energy stored and recovered by the muscular-hydrostat fin, and estimated energy output from work (E = F·d) since no intake-versus-output data exist for fin propulsion alone.
- Proposed validation experiments: a soft biomimetic cuttlefish robot in a flow tank with force transducers, PIV, and strain gauges, and a live-animal flow-tank test to measure fin thrust using Newton's second law.

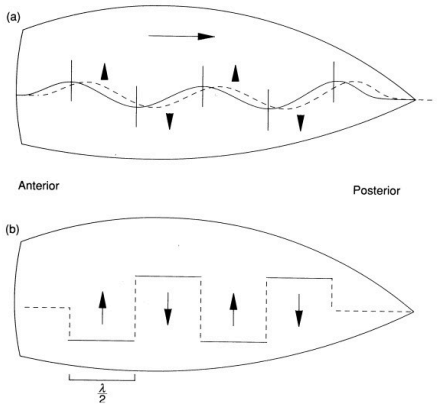


Fig. 2. Sinusoidal wave travelling from anterior to posterior along the fin (Kier et al.).

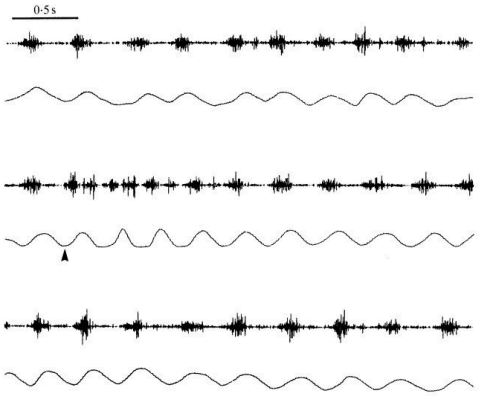


Fig. 3. EMG recordings showing fin muscles firing in step with the fin beat (Kier et al.).

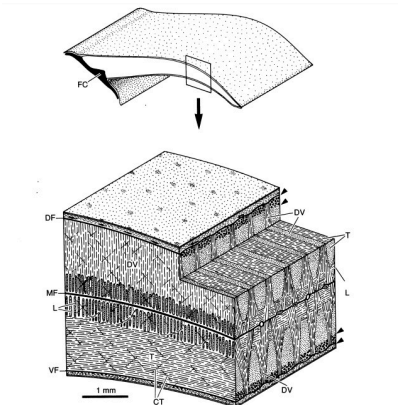


Fig. 4. Muscular-hydrostat fin: dorsoventral, transverse, and longitudinal muscle layers (Kier, 1989).

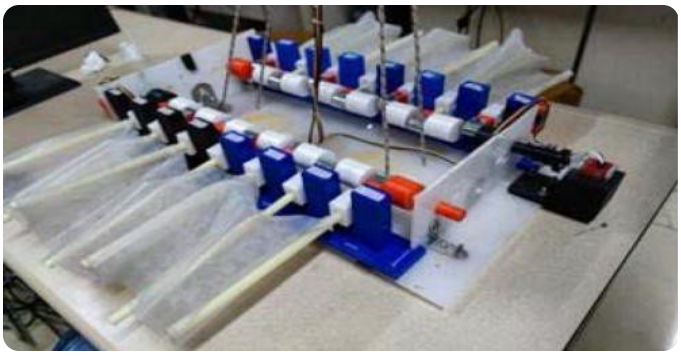


Fig. 5. Robotic flexible-fin model used for the robotic side of the hybrid (Su et al., 2022).

SOFT ROBOTICS · 2025 · SENIOR PROJECT · CASE WESTERN RESERVE UNIVERSITY · ADVISOR: PROF. CHASE CAO

Liquid Dielectric Elastomer Actuator Motor for Underwater Propulsion

Jeffrey Kagle, Katharine Lobas, Jarod Lau, Yi Jin, Chase Cao

Rigid motors and hydraulics struggle under deep-water pressure and tend to be bulky and inefficient, while conventional dielectric elastomer actuators deliver high force but little displacement. We designed a deformable rotary motor driven by liquid dielectric elastomer actuators (Peano-HASELs), selected the concept through weighted trade studies, and built custom quarter-circle actuators and a 3D-printed motor housing as a proof of concept.

↓ Final Report ↓ Poster · CWRU Intersections 2025

4
liquid Peano-HASEL actuators pulling in sequence on one rotor

20 cm
motor diameter, 3D-printed frame on two ball bearings

5–9 kV
drive voltages tested; actuators deformed without damage

3.1
trade-study score for rotary motion (flapping 2.45, jetting 2.2)

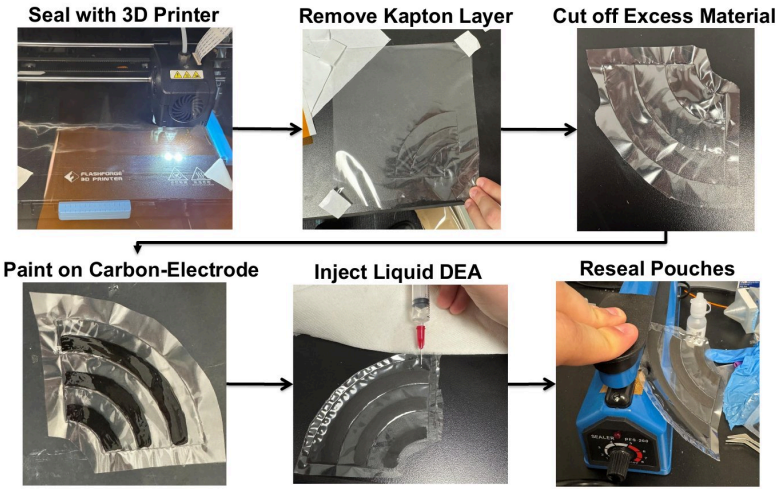


Fig. 1. Actuator fabrication: heat-seal BOPP film with a 3D printer, remove the Kapton layer, trim, paint carbon electrodes, inject the liquid dielectric, and reseal the pouches.

KEY RESULTS

- Weighted trade studies selected liquid DEAs as the actuation method (score 2.9, ahead of solid DEAs at 2.7, motors at 2.6, and hydraulics at 2.0) and rotary motion as the movement method (3.1 vs. 2.45 for flapping and 2.2 for jetting).
- Designed a three-pouch, quarter-circle Peano-HASEL actuator with painted carbon electrodes so four units fit around a circular frame and pull on an eccentric central rotor.
- Developed the fabrication process: heat-sealing BOPP film with a 3D printer, painting carbon electrodes, injecting silicone oil as the liquid dielectric, and resealing the pouches.
- Fabricated actuators deformed at 5, 8, and 9 kV from an off-board high-voltage DC/DC converter without damage; testing was limited to verifying functionality.
- Identified the main manufacturing fixes for future builds: thicker seal lines in the printer path to prevent leaks, better-placed injection points, and fewer air bubbles in the pouches.

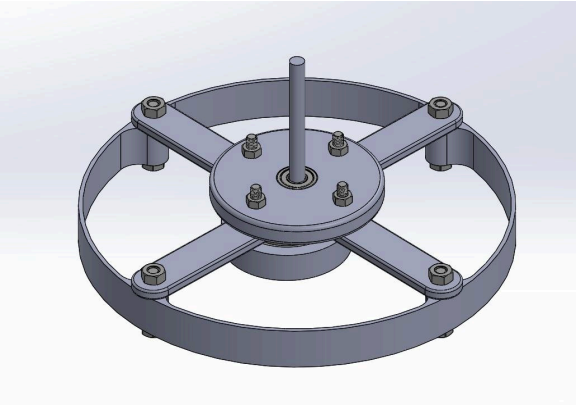


Fig. 2. Full CAD assembly: 3D-printed frame, support struts, and central hub with shaft on two ball bearings.

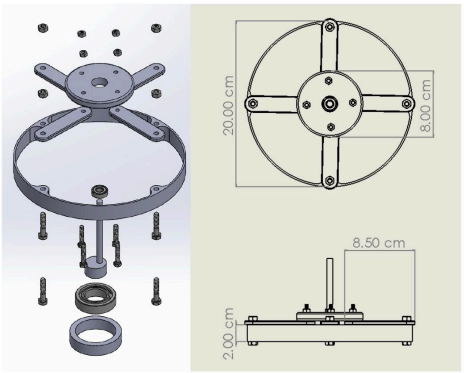


Fig. 3. Exploded view and dimensioned drawing of the 20 cm motor.

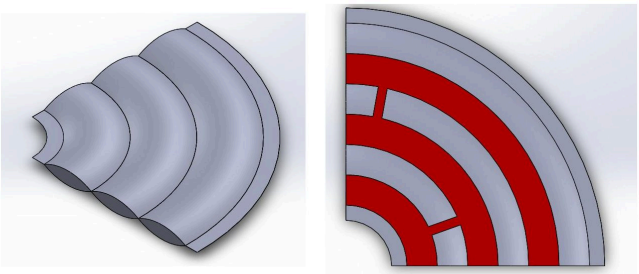


Fig. 4. Quarter-circle actuator design (left) and painted carbon electrode layout (right).

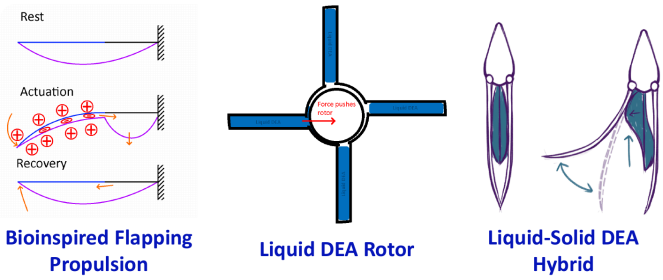


Fig. 5. Concepts compared in the trade studies: bioinspired flapping propulsion, liquid DEA rotor, and liquid–solid DEA hybrid.

PRODUCT DESIGN · 2026 · MAE 5630 ADVANCED PRODUCT DESIGN · CORNELL

ANCORE U Tiller Extension Joint

Mae Sliwinski, Garrett Smith, Alex Jenkins, **Jeffrey Kagle**

Competitive sailing hardware with a three-layer elastomeric architecture — polyurethane body, nylon rope core, and woven polymer mesh sleeve. Patent pending.

↓ Product Design Specification



Fig. 1. CAD renderings of the tiller extension (left) and the mesh-sleeved joint (right).